Do Sea-Level Rise Maps Backfire? A California-Focused Causal Tutorial

AUTHORS
Joshua Ferrer-Lozano
Zach Loo
William Mullins
Sofia Sarak

PUBLISHED
March 6, 2026

Background

Sea-level rise (SLR) is one of California's most pressing climate risks. Yet even in a climate-aware state, public concern about SLR remains surprisingly low. Mildenberger et al. (2024) test whether personalized SLR maps increase resident concern by showing how rising seas could affect their home.

The surprising finding: maps often reduce perceived personal harm, even for people whose homes are projected to flood.

This blog post walks through the design-based causal method used in the study and recreates the California-only version of Figure 4, which shows how the map's treatment effect varies with distance from the SLR boundary. The original study conducted this analysis on coastal communities of New Jersey, Florida, and Virginia, as well. California data comes from the San Francisco Bay Area.

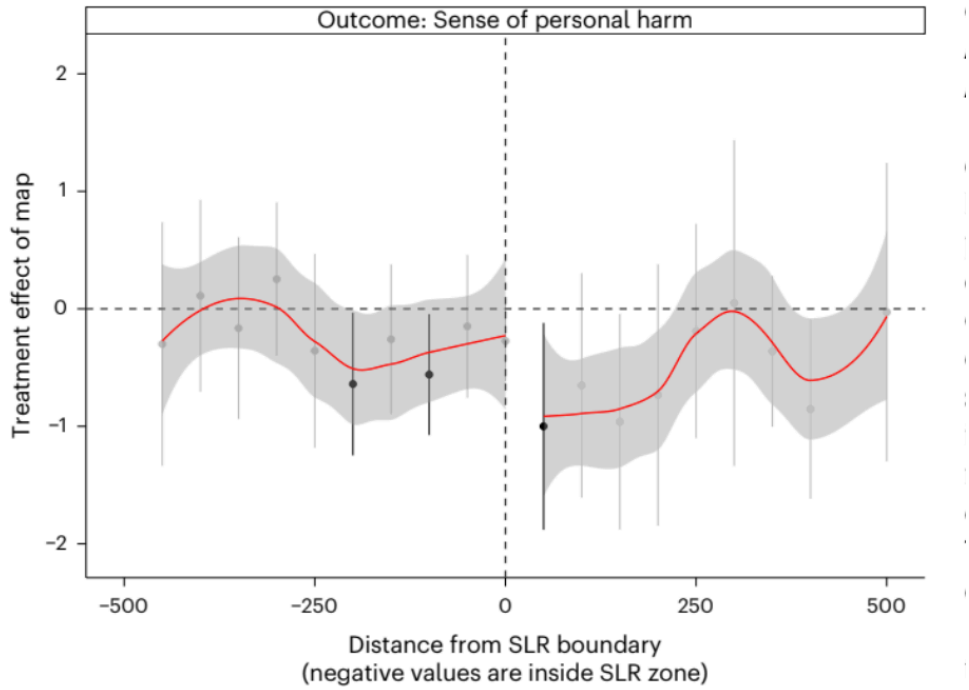


Figure 4 (from Mildenberger et al., 2024): Effect of map treatment on personal concern, with separate regressions estimated at 50 m intervals, pooled across California, New Jersey, Virginia and Florida samples.

Study Design

Randomized experiment

Respondents in California were recruited using address-level mail-to-web sampling. Each respondent was randomly assigned to:

Treatment (treat_map = 1): See a customized SLR map centered on their home

Control (treat_map = 0): See no map

Because treatment is randomized, the difference in outcomes between treated and control respondents is meant to be a causal estimate of the map's effect.

Additionally, the study utilized a geographic risk discontinuity design to test whether there was a difference in concern between respondents that were just within or just outside of the flood zone. Under these assumptions, these residents should be relatively similar in most covariates other than the risk to sea level rise. A jump in concern between groups on either side of the threshold would indicate a treatment effect.

Survey Methods

Individualized letters were sent to households within the sample area (San Francisco Bay, CA) in Fall 2020; the timing of survey completion depended on when respondents chose to complete it. The average response rates inside and outside the projected SLR boundary (as a fraction of mailed invitations) – of all 4 states in the original study – was 2.6% and 3.1% respectively. Treatment, or whether the letter/survey was accompanied by a map of SLR, was randomized using the Qualtric web platform.

Although the randomization assumption was met in this case, the low rate of response has the potential to contribute non-response/participation bias in the outcome.

Outcome variable

`dv_personal` — The study measures perceived risk or personal harm from SLR using a 0-3 scale, which respondents selected in their surveys.

Running variable

To recreate Figure 4, we use:

`dist_min` — distance (in meters) from the SLR boundary, where negative values correspond to homes inside the projected flood zone and positive ones outside of it.

This running variable is used to estimate spatial heterogeneity in the treatment effect.

Causal Identification Strategy

The randomized sample estimates an Average Treatment Effect (ATE). Because map exposure is randomly assigned, the causal effect is simply:

ATE = mean outcome of those that received the SLR map - mean outcome of those that did not receive map

Spatial heterogeneity/geographic risk discontinuity (Figure 4) is leveraged to understand whether the map works differently for people closer to the flood zone. The study does the following:

1. Bin respondents into 50-meter intervals of distance to the SLR boundary
2. Estimate the treatment effect within each bin
3. Plot those bin-specific effects against distance

To reiterate, this is not a formal regression discontinuity design, but it borrows the intuition that if the map increases personal relevance, the treatment effect should be strongest near the boundary.

Study Result

The treatment effect is negative (reduces concern of SLR risk), regardless of geographic location.

Recreation of Results/Figure 4

Load Required Packages

```
library(tidyverse)
library(broom)
library(estimatr)
library(gtsummary)
```

Load and Clean California Data

```
# CA survey data
data_ca <- read_csv("data/SeaLevelRiseSurvey_California_January+16,+2021_13.54.csv") %>%
  .[-c(1,2),] %>%
  mutate(geo = "CA")

# Load CA distance-to-boundary file
dist_ca <- read_csv("data/ca_dist_mins.csv")

# Load CA combined metadata
```

```
cw_ca <- read_csv("data/ca_combined_final_withtraffic_sub.csv")

# Load flood zone file
fld_zone <- read.csv("data/fld_zone.csv")
```

Merge Survey, Distance and Flood Zone

```
dist_ca_clean <- dist_ca %>%
  # Keep only the first occurrence
  filter(!duplicated(.)) %>%
  group_by(adds) %>%
  # Filter to minimum at each address
  filter(dist_min == min(dist_min)) %>%
    inner_join(cw_ca %>%  # Join traffic data on addresses, and retain corresponding emai
      select(adds, email, FID) %>%
      filter(!duplicated(.))) %>%
  rename(RecipientEmail = email) %>%
  inner_join(fld_zone %>% # Join flood zone info on addresses, keeping county, tract, and
             select(adds, COUNTYFP, TRACTCE, FLD_ZONE) %>%
             distinct(adds, .keep_all = TRUE))
```

```
Joining with `by = join_by(adds)`
Joining with `by = join_by(adds)`
```

In the analysis we are replicating in this blog post, traffic data is not used (but it used for another experiment in the same study, which is why it is included here).

Data Description: This intermediary clean data frame contains the following columns:

- `adds` - Residential addresses where surveys were sent
- `dist_min` - The minimum distance to the SLR boundary
- `RecipientEmail` - Email address associated with recipient of survey
- `FID` - Feature IDs from the corresponding geospatial data set
- `COUNTYFP` - County identifier (there are 10 unique counties included in this study)
- `TRACTCE` - Census tract identifier
- `FLD_ZONE` - Flood zone area identifier

Construct California-Only Analytic Dataset

```
# Create function that recodes survey responses to numeric values
dv_recode <- function(x){
```

```
  case_when(
    x == "Not at all" ~ 0,
    x == "Only a little" ~ 1,
    x == "A moderate amount" ~ 2,
    x == "A great deal" ~ 3
  )
}

# Clean CA data frame
data_ca_clean <- data_ca %>%
  # Join with cleaned distance data, by email
  left_join(dist_ca_clean, by = "RecipientEmail") %>%

  mutate(
    # Boolean variable denoting whether observation received a map treatment
    treat_map = !is.na(exp_emphasis),

    # Boolean variable denoting whether the image shown was a SLR image
    slr = str_detect(img, "SLR"),

    # Recode survey responses using function made above
    dv_personal = dv_recode(slharmpersonal),

    # Flip sign of dist_min so that negative values represent closer to coast/greater ris
    dist_min = ifelse(slr == TRUE, -dist_min, dist_min)
  ) %>%

  # Remove NA values
  filter(!is.na(dv_personal), !is.na(dist_min))
```

California-Only Covariate Balance Table

```
# Create balance table comparing groups that did and did not receive treatment
data_ca_clean |>
  select(race, edu, gender, ideo, party, dist_min, treat_map) |>
  tbl_summary(
    by = treat_map,
    statistic = list(all_continuous() ~ "{mean} ({sd})")) |>
  modify_header(label ~"**Covariate**") |>
  modify_spanning_header(c("stat_1", "stat_2") ~ "**Map Treatment**")
```

	Map Treatment	
Covariate	**FALSE** N = 356[1]	**TRUE** N = 324[1]
race		
African American	7 (2.1%)	3 (1.0%)
African American,Native American	1 (0.3%)	0 (0%)
African American,Pacific Islander	0 (0%)	1 (0.3%)
Asian	82 (25%)	89 (29%)
Asian,Pacific Islander	1 (0.3%)	0 (0%)
Asian,Pacific Islander,Other	0 (0%)	1 (0.3%)
Hispanic	18 (5.5%)	10 (3.3%)
Hispanic,Native American	0 (0%)	1 (0.3%)
Native American	0 (0%)	1 (0.3%)
Other	9 (2.8%)	11 (3.6%)
Pacific Islander	2 (0.6%)	0 (0%)
White/Caucasian	188 (58%)	169 (56%)
White/Caucasian,African American	2 (0.6%)	3 (1.0%)
White/Caucasian,African American,Hispanic,Asian,Native American,Pacific Islander,Other	0 (0%)	1 (0.3%)
White/Caucasian,Asian	1 (0.3%)	4 (1.3%)
White/Caucasian,Hispanic	8 (2.5%)	6 (2.0%)
White/Caucasian,Hispanic,Asian	2 (0.6%)	1 (0.3%)
White/Caucasian,Native American	4 (1.2%)	1 (0.3%)
White/Caucasian,Other	1 (0.3%)	0 (0%)
Unknown	30	22

[1] n (%); Mean (SD)

	Map Treatment	
Covariate	**FALSE** N = 356[1]	**TRUE** N = 324[1]
edu		
2-year College Degree	20 (6.1%)	15 (4.9%)
4-year College Degree	115 (35%)	98 (32%)
Doctoral Degree	23 (7.1%)	35 (12%)
High School / GED	5 (1.5%)	10 (3.3%)
Less than High School	4 (1.2%)	0 (0%)
Masters Degree	108 (33%)	92 (30%)
Professional Degree (JD, MD)	20 (6.1%)	27 (8.9%)
Some college	30 (9.2%)	25 (8.2%)
Trade school	1 (0.3%)	2 (0.7%)
Unknown	30	20
gender		
Female	116 (36%)	131 (43%)
Male	208 (64%)	172 (57%)
Unknown	32	21
ideo		
Conservative	22 (6.8%)	18 (5.9%)
Extremely conservative	2 (0.6%)	2 (0.7%)
Extremely liberal	16 (4.9%)	19 (6.3%)
Liberal	101 (31%)	105 (35%)
Moderate; middle of the road	89 (27%)	86 (28%)
Slightly conservative	29 (8.9%)	18 (5.9%)
Slightly liberal	66 (20%)	56 (18%)
Unknown	31	20

[1] n (%); Mean (SD)

	Map Treatment	
Covariate	**FALSE** N = 356[1]	**TRUE** N = 324[1]
party		
Democrat	186 (57%)	183 (60%)
Independent	83 (26%)	79 (26%)
Other Party	26 (8.0%)	17 (5.6%)
Republican	30 (9.2%)	24 (7.9%)
Unknown	31	21
dist_min	407 (737)	335 (759)

[1] n (%); Mean (SD)

Data Description: The final clean data frame contains a total of 123 columns, we will not summarize in entirety here. Those that are necessary for our analysis include:

- `dv_personal`: Survey result, recoded as 0-3 to represent responses "Not at all", "Only a little", "A moderate amount", and "A great deal", respectively, when residents were asked about SLR concern. This is our **outcome variable**.
- `treat_map`: Whether treatment was applied (SLR map included in survey). This is our **treatment variable**.
- `dist_min`: Minimum distance (in meters) to SLR boundary. This is the **running variable** for our spatial heterogeneity test.

This data contains 680 observations from California residents. The entire dataset (not used here) containing data from all four states has a sample size of 1,243

Step 1 - Estimate the ATE of California

```
# Run robust linear model, using personal risk as response and map treatment as predictor
mod_ate <- lm_robust(dv_personal ~ treat_map, se_type = "HC1",
                     data = data_ca_clean)

# Print summary
summary(mod_ate)
```

```
Call:
lm_robust(formula = dv_personal ~ treat_map, data = data_ca_clean,
    se_type = "HC1")
```

```
Standard error type:  HC1

Coefficients:
              Estimate Std. Error t value   Pr(>|t|) CI Lower CI Upper  DF
(Intercept)     1.5646    0.04869  32.137 2.129e-138   1.4690   1.6602 678
treat_mapTRUE  -0.2529    0.07454  -3.392  7.332e-04  -0.3992  -0.1065 678

Multiple R-squared:  0.01685 ,  Adjusted R-squared:  0.0154
F-statistic: 11.51 on 1 and 678 DF,  p-value: 0.0007332
```

Step 2 — Create distance bins

```r
# Set bin width
bin_width <- 50

# Create data set where each observation is binned based on distance away from SLR line
data_ca_binned <- data_ca_clean %>%
  mutate(
    dist_bin = bin_width * round(dist_min / bin_width),
    dist_bin = pmax(pmin(dist_bin, 500), -500)
  )
```

Step 3 - Estimate treatment effect within each bin

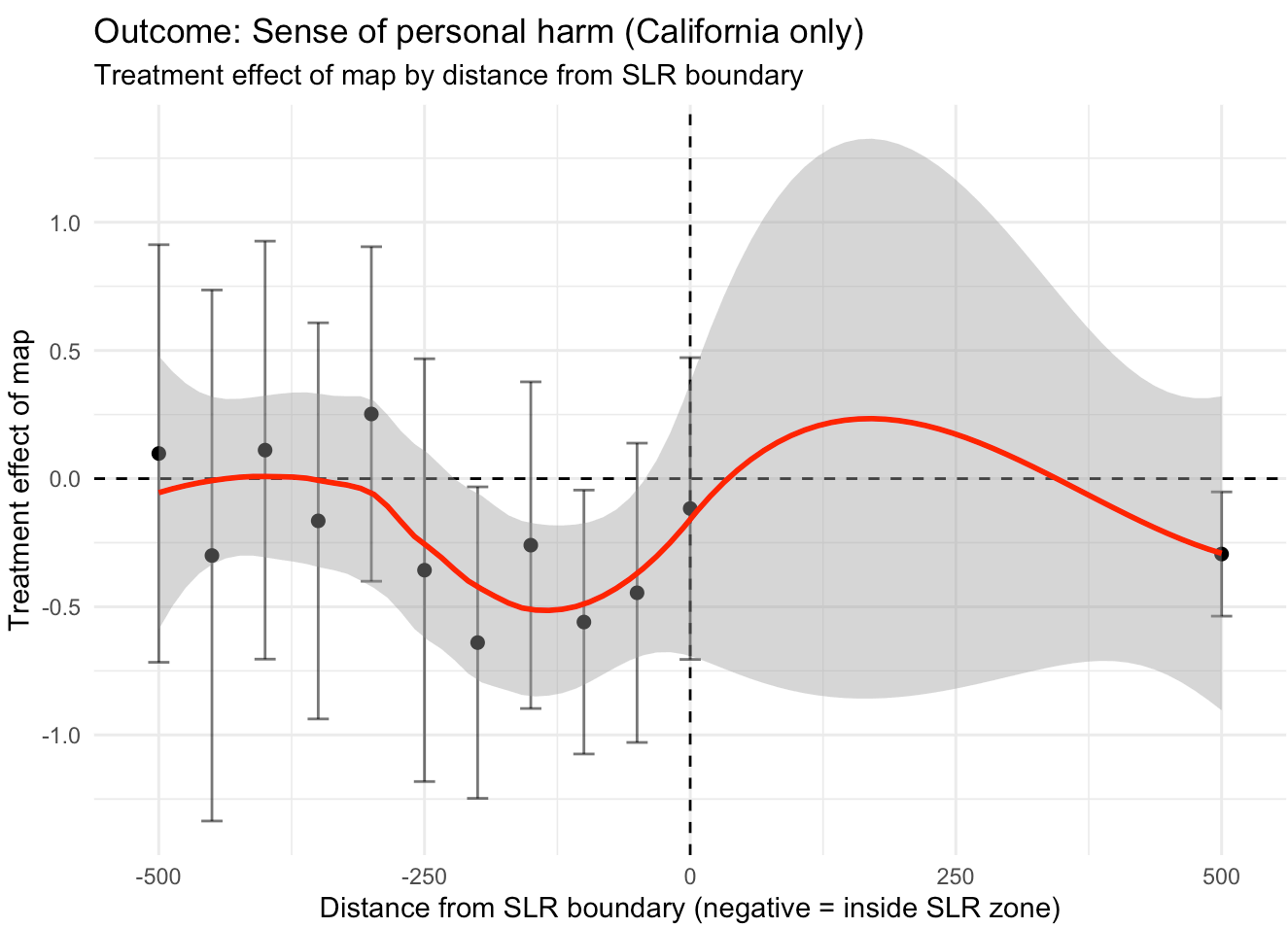
```r
effects_ca <- data_ca_binned %>%

  # Keep only distance bins that are more than 15 meters away from SLR line/flood zone
  group_by(dist_bin) %>%
  filter(n() >= 15) %>%

  # Run same model as before
  do(tidy(lm_robust(
    dv_personal ~ treat_map,
    se_type = "HC1",
    data = .
  ))) %>%

  # Ungroup
  ungroup() %>%

  # Keep only observations that received treatment
  filter(term == "treat_mapTRUE") %>%

  # Calculate confidnece intervals
  mutate(
    ci_low = estimate - 1.96 * std.error,
```

```
    ci_high = estimate + 1.96 * std.error
  )
```

Step 4 - Plot California- Only Figure

```
# Replicate Figure 4, only with SF Bay Area data
ggplot(effects_ca, aes(x = dist_bin, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high), width = 20, alpha = 0.6) +
  geom_smooth(method = "loess", se = TRUE, color = "red") +
  theme_minimal() +
  labs(
    title = "Outcome: Sense of personal harm (California only)",
    subtitle = "Treatment effect of map by distance from SLR boundary",
    x = "Distance from SLR boundary (negative = inside SLR zone)",
    y = "Treatment effect of map"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Interpretation

Three key insights emerge from the California analysis:

1. **Maps reduce perceived personal harm across almost all distances.** Even respondents whose homes are projected to flood show lower concern after seeing the map.
2. **There is no sharp change at the SLR boundary.** If maps increased personal relevance, we would expect a spike near 0 meters. Instead, the curve is flat and negative.
3. **California behaves like the national sample.** Despite higher climate awareness, the backfire effect persists. This insight is based on evidence from the original study, as well.

Conclusion and Summary

This study utilizes a randomized experiment to produce causal inference results. Additionally, spatial heterogeneity allows for mechanism exploration. However, due to the survey methodology, non-response bias persists.

For policymakers, the takeaway is clear: SLR maps may unintentionally reduce concern, even among at-risk households.

Causal Identification:

- The study's identification strategy is a randomized survey experiment that provides credibility.
- In the California sample, the treatment assignment (treat_map) is independent of respondent characteristics because the map is randomly shown after the individuals enter the survey.
- For ATE and bin-level estimates in Figure 4, the causal effect can be identified if: Randomization was implemented correctly If one respondent's treatment does not affect anther's outcome (SUTVA) This leads to the belief that internal validity is strong and that the randomized design credibly identifies the short-run causal effect of seeing the map on the survey outcome.

Statistical Assumptions:

Because this is a randomized experiment, the assumptions are:

Randomized validity

The author reports no evidence of imbalance across the treatment and control groups Our California-only replication relies on the same randomization

SUTVA

Respondents complete surveys individually and therefore cannot see each others maps Spillovers are unlikely due to the online setting Due to this, we believe that statistical assumptions are reasonable.

Sampling and External Validity:

- The sample are the California respondents that were recruited via “address-matched mail-to-web sampling”
- This generalizes respondents that are Californian (more likely climate aware), metropolitan and more likely to engage with mailed surveys
- Even inside the flood zone, respondents show low personal concern
- These results may not generalize other people that are: Renters Low-income People that dont respond to mail surveys Other states or countries
- Our replication focuses on California which is more climate-concerned than the national-average Because of this, we find external validity is weak and may not be able to be generalized to the broader population.

Therefore, due to the relatively small sample size and the geographic limit of only one coastal community, it can be concluded that these results can only be generalized to the residents of the San Francisco Bay Area, California.

Outcome Validity Concerns:

The outcome can be a concern as a single 4-point Likert item asking how much sea level rise will harm them personally raises some issues. The paper mentions respondents spent a median time of 33 seconds on the map page which may introduce a reaction that does not internalize the information fully The lower scores may reflect reassurance (they are not in the flood zone) or disbelief in the map Treatment Validity Concerns

A map of sea level rise shown once at the discretion of the respondents may not capture the full opinion of the respondent. 19% of the respondents found the map not very or not at all credible. We can say that the Measurement validity is decent, however maybe limited in measuring risk perception.

Other Limitations:

Spillover effect

It would not have been possible for the researchers to control whether or not residents discussed the surveys and their responses with each other. This could lead to observations that are not truly independent.

Time lag

Differences in response time (or even allowing for a long response time) could potential skew outcomes. For example, if there was a weather event that led to minor flooding in the time period that the survey was open, respondents who completed the survey before and after the event could be basing their beliefs off of different realities (and not off of the treatment). These discrepancies were not controlled for in the study methods.

Overall, our California-only replication supports the paper's core experimental finding: randomized exposure to personalized SLR maps causally reduces reported personal concern about sea-level rise. The internal validity of this estimate is strong due to proper randomization, minimal risk of spillovers. However, the broader policy relevance of this effect is limited. The outcome measure uses coarse scores and maybe be sensitive to short-run reactions, the treatment is a one-time map rather than the standard policy communication, and the California/San Francisco sample may and probably is unrepresentative of the broader population. These factors weaken external validity and can raise questions about whether the observed "backfire effect". In all, the experiment credibly identifies a short-run causal effect in this specific survey context, but the evidence is less convincing as support for the stronger policy claim that SLR maps broadly reduce public concern in real-world settings.